QUANT

Algebra - 1

(Linear, Quadratic Equations & other imp. concepts)

Assignment 04

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1. Baichung's father is 26 years younger than Baichung's grandfather and 29 years older than Baichung. The sum of the ages of all three is 135 years. What is Baichung's age?

1. 46 2. 17 3. 18 4. 20

2. One of the two digits of a two digit number is three times the other digit. If you interchange the digits of this two-digit number and add the resulting number to the original number, you get 88. What is the half of the original number?

1. 26 2. 13 3.31 4.39

3. Half of a herd of deer are grazing in the field and three fourths of the remaining are playing nearby. The remaining 9 are drinking water from the pond. Find the number of deer in the herd.

1. 83 2. 77 3.72 4. 92

4. One says, "Give me a hundred, friend! I shall then become twice as rich as you". The other replies, "If you give me ten, I shall be six times as rich as you". Tell me what is the amount of their (respective) capitals?

1. 175, 50 2. 40, 170 3. 120, 40 4. 50,175

5. The area of a rectangle gets reduced by 9 square units, if its length is reduced by 5 units and breadth is increased by 3 units. If we increase the length by 3 units and the breadth by 2 units, the area increases by 67 square units. Find the dimensions of the rectangle.

1. 27, 13 2. 25, 14 3. 17, 9 4. 31, 11

6. The ages of two friends Ani and Biju differ by 3 years. Ani's father Dharam is twice as old as Ani and Biju is twice as old as his sister Cathy. The ages of Cathy and Dharam differ by 30 years. Find the ages of Ani.

1. 38 2. 8 3. 19 4. 16

7. There are two consecutive odd positive integers, the sum of whose squares is 290. What is the square of the smaller number?

1. 121 2. 169 3. 225 4. 289

8. How many liters of pure alcohol should be added to 40 liters of 10 % alcohol solution, to obtain a mixture which is 20 % alcohol?

1. 5 2. 20 3. 4 4. 30

9. If α and β are the roots of the equation $x^2 + ax + b = 0$, where $b \neq 0$, then the roots of the equation $bx^2 + ax + 1 = 0$ are

1. $1/\alpha, 1/\beta$ 2. $\alpha^2\beta^2$ 3. $1/\alpha^2, 1/\beta^2$ 4. $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

10. If $4x^{10} - x^9 - 3x^8 + 5x^7 + kx^6 + 2x^5 - x^3 + kx^2 + 5x - 5$, when divided by $(x + 1)$ gives a remainder of -14, then the value of k equals

1. 2 2. 0 3. 7 4. – 2

Answer key & Explanations

1.	2	Let F be the father's age, G be the grandfather's age and B be Baichung's age F = G – 26 and F = B + 29 Also, B + F + G = 135 ⇒ F – 29 + F + F + 26 = 135, Solving we get F = 46. So B = 17 years OR going by options, If 17 is Baichung's age, then his father is 17 + 29 = 46 and grandfather is 46 + 26 = 72. The sum of their ages comes out to be 135.
2.	2	From the options, if we double 13 we get 26 as in the question half of the number is asked, Interchanging the digits of 26, we get 62 and if we add the original no. and the interchanged no. we get 88. Hence the original number is 26 and its half is 13.
3.	3	Going by options if 72 is the total number of deer in the herd, then half i.e 36 are grazing in the field. Remaining = 36. 3/4 of 36 = 27 So total deer in the herd = 36 + 27 + 9 = 72.
4.	2	Checking by options, if the second person gives Rs. 100 to first, then first will have Rs. 140 which is double of Rs.70 that will be left with the second person. On the other hand, if the first person gives Rs. 10 to the second, the second will have Rs. 180 which is six times of Rs. 30, which the first person will have.
5.	3	Let 17 be the original length and 9 be the breadth. So original area = 153 sq. units. Now if its length is reduced by 5 units and breadth is increased by 3 units, then dimensions become l = 12 and b =12. So area = 144, which is 9 less than the original area. Also, if we increase the length by 3 units and the breadth by 2 units, the new dimensions become l = 20 and b = 11, so new area = 220, which is 67 sq units more than the original area.
6.	3	Given A – B = 3; D = 2A ; B = 2C and D – C = 30. Solving we get A = 19 Or simply going by options, if the age of A is 19 then A's father D = 38yrs. Now B can be either be 22 or 16years old. Now Cathy accordingly will be either 11 years or 8 years. Taking B as 16 years and C as 8 years, we get difference between C and D as 30 years.
7.	1	Since the square of the smaller number is asked, we will take the square root of the values given in the options and check . Taking 121, we get 11 as the smaller no. and 13 would be the other number . Taking the squares of these and adding we get 121 + 169 = 290.
8.	1	Let x liters of alcohol be added. $4 + x = \frac{1}{5}(40 + x)$ 20 + 5x = 40 + x ⇒ 4x = 20 x = 5
9.	1	From the given equation, it can be derived that $\alpha + \beta = -a$. Similarly the value of $\alpha\beta = b$.

		Now for the second given equation sum of roots = - a / b and product of the roots = 1/b. As we know the values of a and b, it can be put to calculate the sum of the roots as [α +β] /$\alpha\beta$, which can be simplified as $\frac{1}{\alpha}+\frac{1}{\beta}$. Similarly the product of the roots will 1/$\alpha\beta$. Just combine the sum and product and find that the parts have to be 1/α and 1/β only. Thus first option is the answer.
10.	2	Putting (x + 1) = 0, the value of x is – 1. Now put this value in the polynomial i.e. $4 \times 1 - (-1) - 3 \times 1 + 5(-1) + k \times 1 + 2(-1) - (-1) + k \times 1 + 5(-1) - 5 = -14$. Solve this $4 + 1 - 3 - 5 + k - 2 + 1 + k - 5 - 5 = -14$ $\Rightarrow 2k - 14 = -14 \Rightarrow 2k = 0$ i.e. $k = 0$. Thus 2nd option is the answer.